Infosys - Coding 6

A strip of N cells must each be painted one of 3 colors (R, G, B). Painting cell i with color c costs cost[i][c]. A streak of color c is a maximal run of consecutive cells all painted c. A streak of length L earns a streak bonus of L × Q (Q is a given constant).

Minimize total painting cost − total streak bonuses. Find the minimum total cost.

Input Format

First line: integer n, the number of cells.

Second line: integer q, the bonus per cell in any same-color streak.

Next n lines: 3 space-separated integers each, representing row i of cost (costs for R, G, B).

Constraints

1 <= n <= 10^5

0 <= q <= 10^4

1 <= cost[i][j] <= 10^4

Output Format

The output contains an integer value

Sample Input 0

```
3
10
5 20 30
50 5 40
100 100 5
```

Sample Output 0

```
-15
```

Explanation 0

n=3, q=10. Cheapest per cell: R(5), G(5), B(5) → paint cost = 15. Bonus = 3 × 10 = 30. Net = 15 − 30 = −15.

Sample Input 1

```
2
0
10 20 30
30 20 10
```

Sample Output 1

```
20
```

Explanation 1

n=2, q=0. Cheapest per cell: R(10), B(10) → paint cost = 20. Bonus = 2 × 0 = 0. Net = 20 − 0 = 20.